CHAPTER 8

Relations and Functions

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8.1 Relations

8.1.1 Definition and representations

Definition 8.1. A *binary relation* from A to B is a subset $R \subseteq A \times B$. We write $a\,R\,b$ for $(a,b) \in R$. When $A = B$, R is a relation *on* A. An *n-ary relation* on $A_1, \ldots, A_n$ is a subset of $A_1 \times \cdots \times A_n$.

Example 8.2. On $A = \{1,2,3,4\}$, the divisibility relation $R = \{(a,b) \mid a \mid b\}$ is $\{(1,1),(1,2),(1,3),(1,4),(2,2),(2,4),(3,3),(4,4)\}$.

A relation admits three equivalent representations: as a *set of ordered pairs*; as a *Boolean matrix* M_R with $(M_R)_{ij} = 1$ iff $(a_i, b_j) \in R$; and, when $A = B$, as a *directed graph* on vertex set A with an arc $a \to b$ for each $(a,b) \in R$. Matrix and graph views turn abstract questions into visual or computational ones: e.g. reflexivity is a diagonal of 1s; symmetry is $M_R = M_R^{\mathsf{T}}$.

Being sets, relations combine by $\cup, \cap, \setminus$; moreover the *inverse* $R^{-1} = \{(b,a) \mid (a,b) \in R\}$ and the *composition* $S \circ R = \{(a,c) \mid \exists b,\ (a,b) \in R \wedge (b,c) \in S\}$ produce new relations from old.

8.1.2 Properties of relations on a set

Definition 8.3. A relation R on A is

- *reflexive* if $\forall a,\ (a,a) \in R$;
- *symmetric* if $(a,b) \in R \Rightarrow (b,a) \in R$;
- *antisymmetric* if $(a,b) \in R \wedge (b,a) \in R \Rightarrow a = b$;
- *asymmetric* if $(a,b) \in R \Rightarrow (b,a) \notin R$;
- *transitive* if $(a,b) \in R \wedge (b,c) \in R \Rightarrow (a,c) \in R$.

Symmetric and antisymmetric are not opposites: the equality relation is both; the relation $\{(1,2),(2,1),(1,3)\}$ *is neither. Asymmetric = antisymmetric + irreflexive.*

Example 8.4. On $\mathbb{Z}$: $\leq$ is reflexive, antisymmetric, transitive; $<$ is asymmetric and transitive; "$a \equiv b \pmod 5$" is reflexive, symmetric, transitive; "a divides b" on $\mathbb{Z}^+$ is reflexive, antisymmetric, transitive.

8.1.3 Equivalence relations

Definition 8.5. An *equivalence relation* is a reflexive, symmetric and transitive relation on A. The *equivalence class* of a is $[a] = \{x \in A \mid x\,R\,a\}$; the set of classes is the *quotient* A/R.

Theorem 8.6 (Fundamental partition theorem)**.** The equivalence classes of an equivalence relation on A form a partition of A; conversely, every partition of A arises this way from exactly one equivalence relation ("belong to the same block").

Proof. Reflexivity puts each a in $[a]$, so classes are nonempty and cover A. Suppose $[a] \cap [b] \neq \varnothing$, say c in both. For any $x \in [a]$: $x\,R\,a$, $a\,R\,c$ (symmetry from $c\,R\,a$), $c\,R\,b$, so by transitivity $x\,R\,b$, i.e. $[a] \subseteq [b]$; symmetrically $[b] \subseteq [a]$. Thus classes are equal or disjoint. The converse verification is routine. $\square$

Example 8.7 (Congruence modulo 3)**.** On $\mathbb{Z}$ define $a \sim b \iff 3 \mid (a-b)$. Reflexive: $3 \mid 0$. Symmetric: $3 \mid (a-b) \Rightarrow 3 \mid (b-a)$. Transitive: $3 \mid (a-b)$, $3 \mid (b-c) \Rightarrow 3 \mid ((a-b)+(b-c)) = a-c$. The classes are $[0] = \{\ldots,-3,0,3,\ldots\}$, $[1] = \{\ldots,-2,1,4,\ldots\}$, $[2] = \{\ldots,-1,2,5,\ldots\}$ — the partition of $\mathbb{Z}$ into residues mod 3, and $\mathbb{Z}/{\sim} = \mathbb{Z}_3$.

8.1.4 Order relations

Definition 8.8. A *partial order* is a reflexive, antisymmetric, transitive relation (e.g. $\subseteq$ on $\mathcal{P}(S)$, $\mid$ on $\mathbb{Z}^+$). It is a *total* order when any two elements are comparable (e.g. $\leq$ on $\mathbb{R}$). A *strict order* (e.g. $<$, $\subset$) is irreflexive and transitive.

8.1.5 Closures

Definition 8.9. The *reflexive* (resp. *symmetric*, *transitive*) *closure* of R is the smallest relation containing R with the stated property: $r(R) = R \cup \{(a,a)\}_{a \in A}$, $s(R) = R \cup R^{-1}$, and $t(R) = \bigcup_{k \geq 1} R^k$, where R^k is the k-fold composition — $(a,b) \in t(R)$ iff the digraph of R has a path from a to b.

8.2 Functions

8.2.1 Definition

Definition 8.10. A *function* $f : A \to B$ is a relation $f \subseteq A \times B$ such that every $a \in A$ is related to *exactly one* $b \in B$, written $b = f(a)$. A is the *domain*, B the *codomain*, and $f(A) = \{f(a) \mid a \in A\}$ the *range* (image).

Useful special cases: the *identity* $\mathrm{id}_A(a) = a$; *constant* functions; the *restriction* $f|_S : S \to B$ of f to $S \subseteq A$; and *extensions*, which enlarge the domain.

8.2.2 Injections, surjections, bijections

Definition 8.11. $f : A \to B$ is

- *injective* (one-to-one) if $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$ — distinct inputs give distinct outputs;
- *surjective* (onto) if $\forall b \in B\ \exists a \in A,\ f(a) = b$ — the range is all of B;
- *bijective* if it is both.

Example 8.12. $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x$ is injective, not surjective. $g : \mathbb{R} \to [0, \infty)$, $g(x) = x^2$ is surjective, not injective. $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^3$ is bijective.

Theorem 8.13 (Finite-case shortcut)**.** Let $|A| = |B|$ be finite and $f : A \to B$. Then f injective $\iff$ f surjective $\iff$ f bijective.

This equivalence — a form of the pigeonhole principle (Chapter 10) — fails for infinite sets: $x \mapsto 2x$ on $\mathbb{Z}$ is injective but not surjective.

8.2.3 Composition

Definition 8.14. The *composition* of $f : A \to B$ and $g : B \to C$ is $g \circ f : A \to C$, $(g \circ f)(a) = g(f(a))$.

Theorem 8.15. Composition is associative. If f and g are injective (resp. surjective, bijective), so is $g \circ f$. Conversely, if $g \circ f$ is injective then f is injective; if $g \circ f$ is surjective then g is surjective.

8.2.4 Inverse functions

Theorem 8.16 (Existence of the inverse)**.** $f : A \to B$ has an *inverse function* $f^{-1} : B \to A$ (with $f^{-1}(f(a)) = a$ and $f(f^{-1}(b)) = b$) if and only if f is a bijection. Moreover $(f^{-1})^{-1} = f$ and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof sketch. If f is bijective, each $b \in B$ has exactly one preimage, and sending b to it defines f^{-1}. If f^{-1} exists, f must be injective (two equal images would need two values of f^{-1}) and surjective (every b equals $f(f^{-1}(b))$). $\square$

8.2.5 Images and preimages of sets

Definition 8.17. For $S \subseteq A$ and $T \subseteq B$: $f(S) = \{f(x) \mid x \in S\}$ and $f^{-1}(T) = \{x \in A \mid f(x) \in T\}$ (defined even when f has no inverse function).

Preimages behave perfectly: $f^{-1}(T_1 \cup T_2) = f^{-1}(T_1) \cup f^{-1}(T_2)$ and likewise for $\cap$ and complements. Images preserve unions, but only $f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2)$ in general, with equality for injective f.

8.3 Cardinality via bijections

Definition 8.18. Sets A and B are *equinumerous*, $|A| = |B|$, when a bijection $A \to B$ exists. A is countable when $|A| \leq |\mathbb{N}|$.

Example 8.19 (A bijection $(0,1) \to \mathbb{R}$)**.** $x \mapsto \tan(\pi x - \frac{\pi}{2})$ maps $(0,1)$ bijectively onto $\mathbb{R}$ (strictly increasing, range $\mathbb{R}$). Hence a bounded interval has *exactly as many* points as the whole real line.

Theorem 8.20 (Cantor)**.** For every set A, $|A| < |\mathcal{P}(A)|$: the diagonal set $D = \{x \in A \mid x \notin f(x)\}$ witnesses that no $f : A \to \mathcal{P}(A)$ is surjective.

Consequently there is an infinite hierarchy of infinities: $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| = |\mathbb{R}| < |\mathcal{P}(\mathbb{R})| < \cdots$

8.4 Summary

Relations are subsets of products, viewable as pair-sets, Boolean matrices or digraphs; the key property bundles are *equivalence relations* (= partitions, Theorem 8.6) and *orders*. Functions are total, single-valued relations; injectivity, surjectivity and bijectivity govern composition (Theorem 8.15) and invertibility (Theorem 8.16); bijections define cardinality, and Cantor's theorem shows the power set always strictly exceeds the set.

Exercises

Exercise 8.1 EASY

Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$. Decide whether R is reflexive, symmetric, antisymmetric, transitive.

Exercise 8.2 EASY

Write the Boolean matrix of $R = \{(a, b) \in A^2 \mid a < b\}$ for $A = \{1, 2, 3, 4\}$, and describe its digraph.

Exercise 8.3 EASY

Which of these functions $\mathbb{Z} \to \mathbb{Z}$ are injective? surjective? (a) $f(n) = n+3$; (b) $g(n) = n^2$; (c) $h(n) = \lceil n/2 \rceil$.

Exercise 8.4 MEDIUM

Prove that congruence modulo m ($m \geq 1$) is an equivalence relation on $\mathbb{Z}$, and determine the number of equivalence classes.

Exercise 8.5 MEDIUM

On $\mathbb{R}^2$, define $(x_1, y_1) \sim (x_2, y_2) \iff x_1^2 + y_1^2 = x_2^2 + y_2^2$. Show $\sim$ is an equivalence relation and describe its classes geometrically.

Exercise 8.6 MEDIUM

Show that divisibility $|$ is a partial order on $\mathbb{Z}^+$ but not a total order. Exhibit two incomparable elements.

Exercise 8.7 MEDIUM

Let $R = \{(1,2), (2,3), (3,1)\}$ on $A = \{1,2,3\}$. Compute the reflexive, symmetric and transitive closures of R.

Exercise 8.8 MEDIUM

Let $f(x) = 3x - 4$ on $\mathbb{R}$. Show f is a bijection and find f^{-1}. Verify $(f^{-1} \circ f)(x) = x$.

Exercise 8.9 MEDIUM

Give an example of $f, g : \mathbb{N} \to \mathbb{N}$ such that $g \circ f$ is injective but g is not, and explain which halves of Theorem 8.15 this illustrates.

Exercise 8.10 HARD

Let $f : A \to B$ and $S_1, S_2 \subseteq A$. Prove $f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2)$, show equality can fail, and prove equality holds for all S_1, S_2 iff f is injective.

Exercise 8.11 HARD

Construct an explicit bijection $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and deduce that $\mathbb{Q}^+$ is countable.

Exercise 8.12 HARD

Using Theorem 8.13, prove: if $|A| = |B| = n$ and $f : A \to B$ is injective, then f is bijective. Then give a counterexample for $A = B = \mathbb{N}$.

Solutions to the Exercises

Solution 8.1. Reflexive: yes — $(1,1), (2,2), (3,3) \in R$. Symmetric: yes — the only non-diagonal pairs $(1,2), (2,1)$ come in mirror pairs. Antisymmetric: no — $(1,2)$ and $(2,1)$ with $1 \neq 2$. Transitive: yes — all required compositions land in R (e.g. $(1,2), (2,1) \Rightarrow (1,1) \in R$). ■

Solution 8.2. $M_R = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ — strictly upper triangular. The digraph has arcs from each smaller number to each larger one and no loops: a transitive tournament-like DAG. ■

Solution 8.3. (a) Bijective: injective since $n+3=m+3 \Rightarrow n=m$; surjective since $n=k-3$ maps to any k. (b) Neither: $g(-1)=g(1)$ kills injectivity; negative integers are not squares. (c) Surjective: every k is $h(2k)$; not injective: $h(1)=h(2)=1$. ■

Solution 8.4. Reflexive: $m \mid 0$. Symmetric: $m \mid (a-b) \Rightarrow m \mid (b-a)$. Transitive: $m \mid (a-b)$, $m \mid (b-c)$ imply $m \mid (a-c)$ (sum). The classes are $[0],[1],\ldots,[m-1]$, one per remainder: exactly m classes. ■

Solution 8.5. The relation compares values of $\varphi(x,y)=x^2+y^2$; "same value under a function" is always reflexive, symmetric, transitive. The class of (a,b) is the circle centred at the origin through (a,b) (the origin alone for radius 0): the plane is partitioned into concentric circles. ■

Solution 8.6. Reflexive: $a \mid a$. Antisymmetric: $a \mid b$ and $b \mid a$ with $a,b>0$ force $a=b$. Transitive: $a \mid b \mid c \Rightarrow a \mid c$. Not total: $2 \nmid 3$ and $3 \nmid 2$, so 2 and 3 are incomparable. ■

Solution 8.7. $r(R)=R\cup\{(1,1),(2,2),(3,3)\}$. $s(R)=R\cup\{(2,1),(3,2),(1,3)\}$. For $t(R)$: paths give $(1,3)$ via 2, $(2,1)$ via 3, $(3,2)$ via 1, then $(1,1),(2,2),(3,3)$ via three-step cycles: $t(R)=A\times A$. ■

Solution 8.8. Injective: $3x-4=3y-4 \Rightarrow x=y$. Surjective: given y, $x=(y+4)/3$ satisfies $f(x)=y$. Hence bijective with $f^{-1}(y)=(y+4)/3$, and $f^{-1}(f(x))=((3x-4)+4)/3=x$. ■

Solution 8.9. Let $f(n)=2n$ and let g map $2k\mapsto k$ and $2k+1\mapsto 0$. Then $(g\circ f)(n)=n$ is injective (indeed bijective), yet $g(1)=g(3)=0$ is not injective. This matches Theorem 8.15: injectivity of $g\circ f$ forces only f to be injective, and surjectivity of $g\circ f$ forces only g to be surjective (g here is surjective). ■

Solution 8.10. If $y\in f(S_1\cap S_2)$, then $y=f(x)$ with $x\in S_1$ and $x\in S_2$, so $y\in f(S_1)$ and $y\in f(S_2)$. Failure: $f(x)=x^2$, $S_1=\{-1\}$, $S_2=\{1\}$: $f(S_1\cap S_2)=f(\varnothing)=\varnothing$ but $f(S_1)\cap f(S_2)=\{1\}$. If f is injective and $y\in f(S_1)\cap f(S_2)$, then $y=f(x_1)=f(x_2)$ with $x_i\in S_i$; injectivity gives $x_1=x_2\in S_1\cap S_2$, so $y\in f(S_1\cap S_2)$. Conversely, if f is not injective, the counterexample pattern above (two points with equal image) breaks equality. ■

Solution 8.11. Cantor pairing: $\pi(m,n)=\frac{(m+n)(m+n+1)}{2}+n$ enumerates the diagonals $m+n=0,1,2,\ldots$ in order; each pair receives a distinct index and every index is reached: a bijection $\mathbb{N}^2\to\mathbb{N}$. Each positive rational is p/q with $(p,q)\in\mathbb{N}^2$ (coprime, $q\geq 1$), so $\mathbb{Q}^+$ injects into $\mathbb{N}^2\sim\mathbb{N}$; being infinite, it is countable. ■

Solution 8.12. If f were not surjective, its range would have at most $n-1$ elements, and $f: A\to f(A)$ with $|A|=n>|f(A)|$ would force two elements to share an image (pigeonhole), contradicting injectivity. Hence f is surjective, so bijective. For $A=B=\mathbb{N}$, $f(n)=n+1$ is injective but misses 0: the shortcut is strictly a finite phenomenon. ■